



[Language Translators]/ [Third]

1. Create a text file to save the following source code.

[First Semester] [2025-2026]

FileName.txt			
[Lab 1]	DEF	[8/10/2017]	[08:30 Am]
Int a,b; "comment"			
Instructor Information	By Info x,y		
Instructor	BODY	Email	Hours
[Bahera Hani]	{	[ruaa.abdullah@nahrainuniv.edu.iq	[2 Hours]
[Ruaa Abdullah]	a=1;	]	
[Ibtihal Abbas]	b=(a*2)/3;		
	y=x*x;		
	y=print;		
	b=print;		
	}		

2. Using an Exceptions: -

An exception is a problem that occurs during program execution. Exceptions cause abnormal termination of the program. Exception handling is a powerful mechanism that handles runtime errors to maintain normal application flow. An exception can occur for many different reasons. Some examples: - A user has entered invalid data. - A file that needs to be opened cannot be found. - A network connection has been lost in the middle of communications. - Insufficient memory and other issues related to physical resources.

Exception Handling:-

Exceptions can be caught using a combination of the **try and catch** keywords. A try/catch block is placed around the code that might generate an exception. Syntax:

```
try {  
    //some code  
} catch (Exception e) {  
    //some code to handle errors  
}
```

A catch statement involves declaring the type of exception you are trying to catch. If an exception occurs in the try block, the catch block that follows the try is checked. If the type of exception that occurred is listed in a catch block, the exception is passed to the catch block much as an argument is passed into a method parameter.

Notice the (Exception e) statement in the catch block - it is used to catch all possible Exceptions.

throw The throw keyword allows you to manually generate exceptions from your methods. Some of the numerous available exception types include the `IndexOutOfBoundsException`, `IllegalArgumentException`, `arithmeticException`, and so on. For example, we can throw an `ArithmeticException` in our method when the parameter is 0.

```
int div(int a, int b) throws ArithmeticException {  
    if(b == 0) {  
        throw new ArithmeticException("Division by
```



[Language Translators]/ [Third]

```
Zero");  
[Fi }  
else {  
[Lab return a / b;  
}  
Inst }  
Instr }
```

[08:30 Am]

Hours

3. Working with Files

The java.io package includes a File class that allows you to work with files. To start, create a File object and specify the path of the file in the constructor.

```
import java.io.File;  
...  
File file = new File("C:\\data\\FileName.txt");
```

Reading Files are useful for storing and retrieving data, and there are a number of ways to read from a files. One of the simplest ways is to use the Scanner class from the java.util package. The constructor of the Scanner class can take a File object as input. To read the contents of a text file at the above path, we would need to create a File object with the corresponding path and pass it to the Scanner object. Reading a File The Scanner class inherits from the Iterator, so it behaves like one. We can use the Scanner object's next() method to read the file's contents.

```
try {  
File x = new File("C:\\data\\FileName.txt");  
Scanner sc = new Scanner(x);  
while(sc.hasNext()) {  
System.out.println(sc.next());  
}  
sc.close();  
} catch (FileNotFoundException e)  
{  
System.out.println("File Not Found"); }
```

The file's contents are output word by word, because the next() method returns each word separately.

It is always good practice to close a file when finished working with it. One way to do this is to use the Scanner's close() method.

4. Needed methods: -

hasNext()
nextLine()
Str.charAt()
Str.Substring()

5. The program: -

```
import java.io.*;  
import java.util.*;  
public class Lexical {  
    public static void main(String[] args) throws Exception{  
        Scanner in;
```

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[Ruaa Abdullah]		
[Ibtihal Abbas]		

```

in = new Scanner(new File("C:\\Users\\User\\Desktop\\nt.txt"));
char ch;
String line, tk, com;
while(in.hasNext()){
    line = in.nextLine();    /*read one line from file*/
    tk = "";
    com = "";
    for(int i=0; i<line.length(); i++){
        ch = line.charAt(i);
        if((ch != ' ') && (ch != "")){
            tk += ch;
        } else if (ch==""){
            com =line.substring(i);
            System.out.println(com);
            com = "";
            i=line.length();
        }
        else{
            System.out.println(tk);
            tk = "";
        }    /*end for loop*/
    }
    System.out.println(tk);
    Tk="";    /*end while loop*/
    in.close(); /*close text file*/
}/*end main method*/
}/*end Lexical class*/
    
```